

AYUSH RAJ

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EXPERIENCE

Software Developer Intern ↗

Horizon Mars Rover Team, CUSAT

📅 Nov 2023–Present 📍 Kochi, Kerala

- Developed and implemented a **URDF**-based robotic arm model with **TF** and **Gazebo** in **ROS 2**, including forward and inverse kinematics.
- Controlled a robotic arm using an **Arduino Due** (32-bit ARM) and servo motors in **ROS 2**, integrating real-time kinematics.
- Technologies used** : **C++**, **ROS 2 (Humble)**, **Linux**, **Arduino IDE**

Embedded Intern ↗

Brainspired Labs Pvt. Ltd. 📍 OnSite, Kochi

📅 Jan 2025 – March 2025

- Integrated a dual 16-bit DAC with **STM32H753ZI** via **I2C**, analyzing output on a digital oscilloscope. Enabled **UART** communication in **ESP32** with an Android app for frequency & voltage control.
- Integrated **CJMCU-1334** audio DAC to **esp32-s3** using **I2S** protocol & gained hands-on experience with **AD5750** eval board.
- Technologies used** : **STM32CUBE-IDE**, **Arduino IDE**, **Embedded C**

PROJECTS

TurtleSim Project ↗



📅 Nov 2024

- Developed a Turtle Catcher game where turtles appear randomly, and the objective is to catch them all.
- Tech Stack**: **Python**, **C++**, **ROS2 (Humble)**

Mobile Robot Simulation ↗



📅 March 2025

- Developed an autonomous mobile robot with a robotic arm using **URDF** in **ROS 2 Humble**, integrating multiple **TFs**, **links**, and **joints**
- Implemented **SLAM** and **Nav2** for autonomous navigation and successfully simulated the robot in **Gazebo**, demonstrating accurate kinematics and motion.
- Tech Stack**: **ROS2 Humble**, **Rviz2**, **Xacro**, **URDF**, **Gazebo 11**, **SLAM**, **NAV2**

Preventive AI ↗



📅 Sept 2024 - Oct 2024

- An AI platform combining **medical imaging**, **genomic data**, and **EHRs** for **early cancer detection** and **personalized treatment insights**.
- Tech Stack**: **Python**, **Supervised ML**, **EJS**, **CSS**, **Node.js**, **Express.js**, **MongoDB Atlas**, **TensorFlow Serving**

DIKSHA ↗



📅 Nov 2024

- Built a full stack application offering free educational tools, including **problem-solving resources**, an **NEP syllabus generator**, **AI chatbot**, **Doc QnA**, **Quiz**, and more.
- Tech Stack**: **Bootstrap**, **Gemini API**, **Render**, **Node.js**, **Express.js**, **MongoDB Atlas**, **Vertex AI**, **LLMware**

SOFT SKILLS

Leadership, Decision Making, Teamwork, Creativity

EDUCATION

Bachelor of Technology, Computer Science

Cochin University Of Science And Technology (CUSAT)

📍 Kochi, Kerala

📅 Oct 2022 – May 2026 (Expected)

Currently in **3rd Year (Sem-6)**- **CGPA** : **9.6**

Academic Achievements

- Represented INDIA** at **European Rover Challenge, 2024, Poland** - Robotics Competition. Team name : **Horizon** (World Rank **18th**)
- Listed among **Top 10% Students** in the Batch.

POSITIONS OF RESPONSIBILITY

- Team Lead** @**HORIZON**- Mars Rover Team, CUSAT
- SIG AI Lead** @**IEEE Execom'24**- CS Branch Society, CUSAT

SKILLS

Languages

Java

Python

Javascript

C/C++

Embedded C

Frameworks and Libraries

React

OpenGL

ROS2

NodeJS

ExpressJS

RestAPI

AXIOS

Gemini API

Gazebo

SLAM

ROS2 NAV2

STM32CUBE - IDE

LLMware

Databases

MongoDB

SQL

Tools

Git/GitHub

Linux

Vertex AI

OOP

AI-ML

Proteus

ACHIEVEMENTS

- Won **5 Hackathons** (out of 10) conducted by **Google Development Group** ↗ (GDG) ,**Major League Hacking** ↗ & **Inovact** ↗.
- Solved **100+** programming problems across various platforms like **LeetCode**, **Hackerrank** etc.
- Jury Choice Award Winner** at **AIC** by **Accenture** & **Finalist** at **DevsHouse'24 Hackathon** by **VIT Chennai**

CERTIFICATIONS

ERC Space & Robotics Competition, AGH University Poland ↗

📅 Sep 2024

ROS2 Course Completion, Udemy ↗

📅 Nov 2024

ROS2 Level 2 Course Completion - Udemy ↗

📅 Feb 2025

Certificate of Appreciation (Tech. Team) - TEDxCUSAT ↗

📅 Sep 2024

Artificial Intelligence Fundamentals, IBM ↗

📅 June 2024